

UDARA KURUKULASOORIYA

Software Engineer | Full-Stack

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PROFILE

Software Engineer with three years of full-stack experience delivering enterprise web applications across the .NET and modern TypeScript ecosystems, from legacy-system modernisation to greenfield platform development. Strong grounding in relational data design and API architecture, with recent hands-on specialisation in applied GenAI and LLM engineering — retrieval-augmented generation, parameter-efficient fine-tuning and multi-agent systems.

WORK EXPERIENCE

Software Engineer | Avonet Technologies

Aug 2025 – Present

Regional IT | ALMADS

- Converting legacy Clarion application logic into a modern C# .NET MVC architecture, including the migration and refactoring of stored procedures and core business logic.
- Developed responsive grid views and reusable UI components using Kendo UI, connecting dynamic in-grid data loading to backend stored procedures through MVC patterns.

Technologies: .NET MVC, C#, Kendo UI, jQuery, SQL Server, Telerik Reporting, Azure DevOps

Kinetix ML Studio — Platform Layer

- Contributed to the shared application platform behind a no-code machine-learning studio, built as a pnpm/Turborepo monorepo with a NestJS API and a Next.js front end.
- Implemented authentication and role-based access control, enforcing a database-driven privilege matrix (CASL) across API routes with Keycloak and Auth.js session handling.
- Built master-data and audit-trail modules using Prisma over PostgreSQL, with Zod schemas shared across backend and frontend to drive validation, typing and API documentation from a single contract.
- Covered platform endpoints with automated integration tests using Jest and Testcontainers.

Technologies: NestJS, Next.js, TypeScript, Prisma, PostgreSQL, Zod, CASL, Keycloak, TanStack Query, Docker

Software Engineer | Sitecore

Sep 2024 – Aug 2025

- Developed and maintained scalable web applications using .NET 7 and Angular, improving overall system efficiency.
- Provided technical support for Moosend, Send, Discover and Sitecore Search, diagnosing and resolving partner implementation issues to enhance product performance.
- Deconstructed and analysed partner implementations to identify and resolve technical issues.
- Participated actively in Agile ceremonies — sprint planning, daily stand-ups and retrospectives — using JIRA for task management.

Technologies: .NET 7, Angular, TypeScript, SQL Server, JIRA

Software Engineer Intern | IFS R&D International (Pvt) Ltd

Dec 2022 – Jul 2023

- Developed and optimised new features for Team ALE using Angular and .NET, ensuring high performance and scalability.
- Enhanced application functionality for better maintainability and efficiency.
- Performed unit testing and participated in peer code reviews to improve code quality and compliance with best practices.

Technologies: Angular, .NET Framework, C#, SQL Server

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, Python, Java, SQL, C

Frameworks & Libraries: .NET 7 / .NET Core / .NET MVC, Angular, Next.js, React, NestJS, Node.js, Spring Boot, Kendo UI, Tailwind CSS, Bootstrap

Databases & Data Access: SQL Server, PostgreSQL, MySQL, MongoDB, Prisma, TypeORM, stored procedures

GenAI & LLM Engineering: RAG design and evaluation (ChromaDB, embeddings, semantic chunking, re-ranking, MRR/nDCG, LLM-as-judge); multi-agent orchestration, tool calling and agent memory; fine-tuning (QLoRA, PEFT/LoRA, 4-bit quantisation, HuggingFace Transformers, PyTorch, Weights & Biases); provider APIs (OpenAI, Anthropic, Gemini, Groq, Ollama, LiteLLM); prompt engineering; Modal serverless GPU; Gradio

Cloud & DevOps: Azure, AWS, Docker, Azure DevOps, GitHub, Git

Tools & Practices: Visual Studio, VS Code, IntelliJ, JIRA, Agile/Scrum, Jest, Playwright, Testcontainers

EDUCATION

BSc. (Hons) in Information Technology — University of Moratuwa, Sri Lanka

Jan 2020 – Dec 2024

- Second Upper-Class Honours (GPA: 3.62).
- Final Year Thesis: "Automated Movie Content Rating System" — an AI-based system for assessing movie content.
- Relevant coursework: Object-Oriented Programming, Data Structures and Algorithms, Data Mining and Data Warehousing, Software Engineering Methods, Database Management Systems, Enterprise Application Development, Network Programming, Cluster Computing, Web Technologies, Computer Networks.

PROJECTS

Agentic AI & LLM Engineering — Self-Directed Engineering Project

2026

- Capstone of an applied LLM engineering curriculum: an autonomous deal-finding system of seven collaborating agents, combining RSS ingestion with schema-validated extraction, RAG-based price estimation over a 400,000-vector Chroma store, a QLoRA fine-tuned Llama 3.2 3B specialist on serverless GPU, a PyTorch regressor, and an ensemble blending all three.
- Curated an 800,000-item training corpus with category rebalancing and versioned splits, and benchmarked every model against a baseline ladder from linear regression and XGBoost up to zero-shot frontier models.
- Built a RAG evaluation harness (MRR, nDCG, LLM-as-judge), then rebuilt the pipeline with semantic chunking, over-retrieval and LLM re-ranking.

Technologies: Python, PyTorch, HuggingFace Transformers, QLoRA/PEFT, ChromaDB, LangChain, Modal, Gradio, Weights & Biases, OpenAI / Anthropic / Gemini APIs

Automated Movie Content Rating System — Final Year Research Project

Sep 2023 – May 2024

- Designed and implemented an AI-driven system to classify movie content against age-appropriateness guidelines.
- Built a deep-learning model using YOLOv8 for human action recognition in video clips; personally developed the sub-module detecting substance abuse.

Technologies: YOLOv8, Python

ALE Portal — IFS Production Project

Dec 2022 – Jul 2023

- Developed a portal allowing users to manage support services and create custom solutions, improving UI flow and backend logic to speed up issue resolution.

Technologies: Angular, .NET Framework

AI PDF Chatbot — Retrieval-Augmented Generation System

2026

- Built a PDF question-answering system from first principles — custom chunking, hybrid vector and BM25 retrieval, rank fusion, re-ranking and page-level citations — instead of an off-the-shelf RAG framework, behind typed protocols that allowed a 45-configuration benchmark sweep without pipeline changes. ~6,500 lines of Python, 113 automated tests.
- Built a retrieval evaluation harness (precision@k, recall@k, MRR, nDCG) over a 29-question ground-truth set to select defaults from measurement; hybrid retrieval raised recall@5 from 0.962 to 1.000, and a controlled 3B/3.8B/20B comparison isolated multi-source conflict detection as a model-capability limit, mitigated with a deterministic detector.

Technologies: Python, Streamlit, ChromaDB, fastembed / ONNX Runtime, Ollama, Google Gemini API, pytest

ShiftReason — Explainable Roster Scheduling System

2026

- Built a nurse and ward shift-scheduling engine on Google OR-Tools CP-SAT that explains its output instead of returning a black-box optimum: infeasible rosters show the conflicting constraints and a minimum-cost repair, and feasible ones include a priced penalty breakdown for every soft-constraint violation.
- Designed as a dependency-free C# domain layer, a guarded solver with model linting, and an ASP.NET Core API streaming live solve progress over SignalR to a React 19 front end, with queued background solving, EF Core/SQLite run persistence and explicit UNKNOWN reporting when feasibility can't be proven; tested with xUnit and Vitest.

Technologies: .NET 10, C#, Google OR-Tools (CP-SAT), ASP.NET Core, SignalR, EF Core, SQLite, React 19, TypeScript, Vite, Tailwind CSS, xUnit, Vitest

Employee Performance Appraisal Application — Second Year Research Project

Dec 2021 – Aug 2022

- Built a web application to streamline the appraisal process and reduce HR workload, specialising in employee rating and performance-based salary increments.

Technologies: Angular, ASP.NET, MS SQL Server

NON-RELATED REFEREES

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